# 1. Introduction

## Background of Case Study

In the heart of Sarawak, the deaf and hard-of-hearing community faces a silent barrier. While global AI focuses heavily on American Sign Language (ASL), **Bahasa Isyarat Malaysia (BIM)**—the linguistic lifeline for thousands in Malaysia—remains digitally underserved. **LinguaSign AI** was born at UNIMAS to bridge this gap, turning standard webcams into real-time interpreters.

## Chosen SDG: Goal 10 – Reduced Inequalities

LinguaSign AI directly addresses **SDG 10** by empowering a marginalized community. By providing a tool that facilitates seamless communication in public and educational sectors, we are reducing the "communication tax" paid by the deaf, ensuring they have equal access to services and social interaction.

## Problem Statement

Current translation methods for the deaf in Sarawak are either **analog** (writing on paper), **expensive** (hiring human interpreters), or **inaccurate** (using ASL-based AI that doesn't recognize BIM). This creates a "data desert" where local signs are ignored by global technology, leading to social isolation and professional barriers.

## The Core Objective

LinguaSign AI is a real-time, two-way translation system designed to bridge the communication gap for the deaf and hard-of-hearing community in Sarawak. It specifically focuses on **Bahasa Isyarat Malaysia (BIM)**, utilizing a **custom-curated dataset** of high-frequency gestures to ensure local linguistic accuracy and practical daily application.

- **Real-World Example**: Imagine a deaf student at UNIMAS trying to ask a librarian a question. Instead of writing on paper, they sign to a laptop camera, and the librarian sees the translation instantly on the screen.
- **The "Sarawak" Factor**: Most AI only knows American Sign Language (ASL). By using local research, we ensure the AI understands Sarawakian BIM nuances rather than "foreign" gestures.

# 2.0 Review of Existing Solutions

To develop a cutting-edge solution, we analyzed current market offerings to identify gaps that LinguaSign AI could bridge.

## 2.1 Competitive Analysis

| Solution | Strengths | Limitations & Gaps |
|---|---|---|
| Global Tech Giants (e.g., Google Gesture) | High financial backing and research depth. | Primarily focused on American Sign Language (ASL); lacks support for local **Bahasa Isyarat Malaysia (BIM)**. |
| Hardware-Based (Sensor Gloves) | Extremely high precision and tracking. | Cost-prohibitive ($500+), cumbersome, and requires hardware that most users don't own. |
| Static Dictionary Apps | Good for learning individual signs. | Does not support real-time conversational translation, because right now requires manual searching. |

## 2.2 The LinguaSign Advantage

Our project fills the "Local Gap." By focusing on a **browser-based, localized BIM solution**, we eliminate the need for expensive hardware while ensuring the AI understands the specific linguistic nuances used in the Malaysian context.

# 3.0 Technical Workflow: The "Pipeline"

## 3.1 Technical Track: The Architecture

LinguaSign AI is built on a "Thin-Client, Heavy-Logic" architecture. We use **MediaPipe** for client-side landmark extraction to ensure privacy and **TensorFlow Lite** for high-speed inference. The system is deployed on a high-performance **VPS (vixdev.cloud)** via **Nginx**, ensuring global accessibility.

## 3.2 The Technical Pipeline (The "Brain" of the System)

The translation process is broken down into five distinct stages to ensure accuracy and speed:

- **A. Input (Capture):** A standard webcam captures a live video stream. This acts as the "eyes" of the system, watching hand motions just like a human interpreter.
- **B. Landmark Extraction (The Skeleton):** Using MediaPipe, the system identifies 21 hand landmarks (joints and fingertips) in 3D space. It sees a digital skeleton, which protects user privacy by ignoring faces and backgrounds.
- **C. Preprocessing (Normalization):** Raw coordinates are scaled so the AI recognizes the sign whether the user is close to the camera or far away. This ensures the system works for all body types and distances.
- **D. Classification (The Neural Network):** A trained **LSTM-based TFLite model** analyzes the motion patterns. It doesn't just look at a still image; it understands the *arc* and *flow* of the sign.
- **E. Output (Translation):** The recognized sign is converted into high-contrast text and displayed on the UI instantly.

### 3.3 Summary Table of Pipeline

| Stage | What happens? | Why it matters? |
|---|---|---|
| **Input** | Webcam takes video. | Low cost; works on any laptop. |
| **Extraction** | 21 dots per hand. | Privacy; the AI sees skeletons, not faces. |
| **Preprocessing** | Normalizing data. | Robustness; works for kids and adults alike. |
| **Classification** | AI "matches" the dots. | Accuracy, optimized for BIM motion patterns. |
| **Output** | Text on screen. | Accessibility bridges the gap for non-signers. |

### 3.4 Non-Technical Track: Design Thinking

We employed a **Human-Centric Design** approach:

1. **Empathize:** Identified the communication barriers faced by the Sarawakian deaf community in public spaces.
2. **Define:** Determined that "latency" (lag) is the biggest enemy of natural conversation.
3. **Ideate:** Selected a web-based deployment to ensure zero-install friction for users.

## 4.0 Results

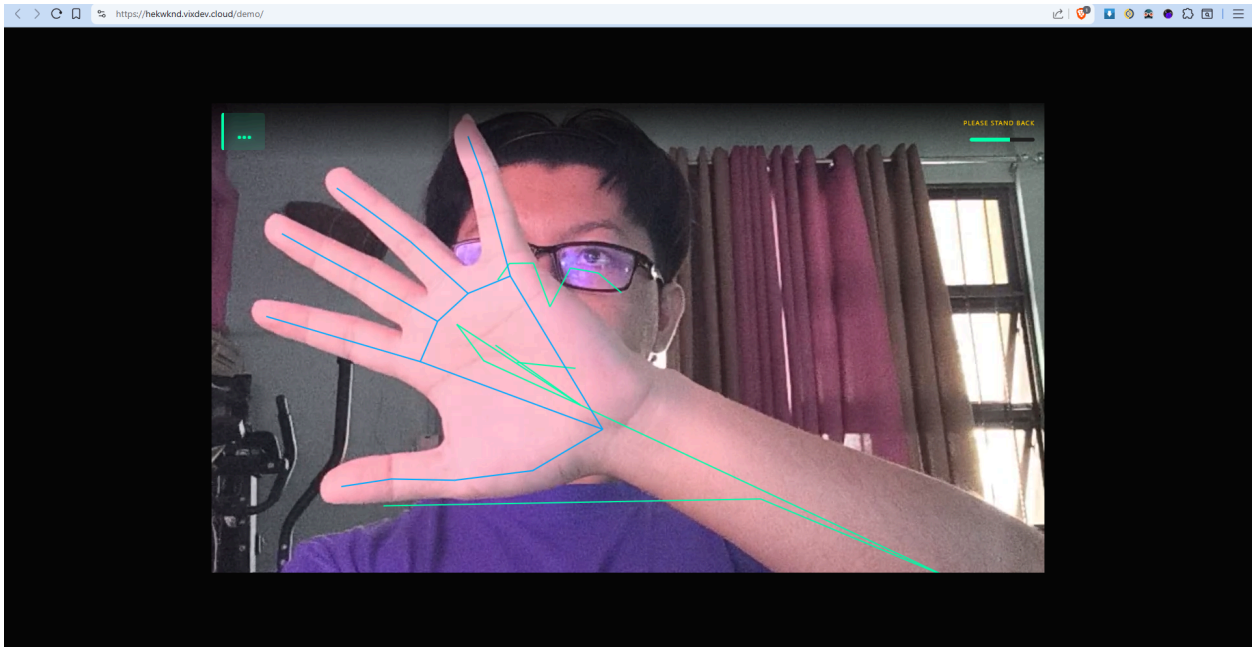
### 4.1 Objectives Achievement

- **Real-Time Translation:** Achieved a processing loop fast enough for conversational flow.
- **Cross-Platform Accessibility:** Successfully deployed on https://hekwknd.vixdev.cloud/demo/ , allowing access via both PC and mobile devices.
- **Localized Vocabulary:** Successfully trained the model to recognize high-frequency daily signs (e.g., "Masalah," "Ribut," "Makan").

### 4.2 Technical Reflection

While testing, we encountered "prediction jitter" on lower-end mobile devices. To maintain a "world-class" user experience, we implemented a **confidence threshold** logic: the UI only updates if the model's certainty exceeds **80%**, preventing flickering or false translations.

## 5.0 Prototype Showcase




- **UI/UX Design:** A minimalist interface focusing on the video feed and the translation box to reduce cognitive load.
- **Real-Time Skeleton Tracking:** High-quality visualization of the 21-point landmark system in action.

## 6.0 AI Acknowledgement

**Disclosure:** This project utilized advanced AI tools to accelerate development and ensure technical rigor.

- **Gemini (Google):** Utilized for system architecture design, Nginx server optimization, and documentation refining.
- **TensorFlow & MediaPipe:** Core AI frameworks used for model training and hand tracking.

# 7.0 Conclusion

LinguaSign AI proves that localized, high-tech accessibility solutions don't require massive budgets or specialized hardware. By leveraging edge computing and **custom BIM-trained models**, we have created a scalable tool for social inclusion.

## Future Roadmap

- **Dataset Expansion:** Training on a wider variety of daily Sarawakian BIM signs.
- **Local Processing:** Moving more "heavy" logic to the user's local device (Localhost/App) to eliminate internet-based lag for a smoother mobile experience.